

Lab 7: Clipping

LAB SHEET

Theory: Explain in brief about Cohen Sutherland line clipping algorithm

Lab works:

Question 1: Write a code in C to implement the Cohen Sutherland line clipping algorithm.

```
#include<graphics.h>

#include<conio.h>

#include<stdio.h>

#include<math.h>

int main()

{

    int rcode_begin[4]={0,0,0,0},rcode_end[4]={0,0,0,0},region_code[4];

    int W_xmax,W_ymax,W_xmin,W_ymin,flag=0;

    float slope;

    int x,y,x1,y1,i, xc,yc;

    int gr=DETECT,gm;

    initgraph(&gr,&gm,"C:\\TURBOC3\\BGI");

    printf("\n***** Cohen Sutherland Line Clipping algorithm *****");

    printf("\n Now, enter XMin, YMin =");

    scanf("%d %d",&W_xmin,&W_ymin);

    printf("\n First enter XMax, YMax =");

    scanf("%d %d",&W_xmax,&W_ymax);

    printf("\n Please enter initial point x and y= ");

    scanf("%d %d",&x,&y);

    printf("\n Now, enter final point x1 and y1= ");
```

```

scanf("%d %d",&x1,&y1);

cleardevice();

rectangle(W_xmin,W_ymin,W_xmax,W_ymax);

line(x,y,x1,y1);

line(0,0,600,0);

line(0,0,0,600);

if(y>W_ymax) {
rcode_begin[0]=1;    // Top

flag=1 ;

}

if(y<W_ymin) {

rcode_begin[1]=1;    // Bottom

flag=1;

}

if(x>W_xmax) {

rcode_begin[2]=1;    // Right

flag=1;

}

if(x<W_xmin) {

rcode_begin[3]=1;    //Left

flag=1;

}

//end point of Line

if(y1>W_ymax){

rcode_end[0]=1;    // Top

```

```

flag=1;

}

if(y1<W_ymin) {
rcode_end[1]=1;      // Bottom
flag=1;
}

if(x1>W_xmax){
rcode_end[2]=1;      // Right
flag=1;
}

if(x1<W_xmin){
rcode_end[3]=1;      //Left
flag=1;
}

if(flag==0)

{

printf("No need of clipping as it is already in window");

}

flag=1;

for(i=0;i<4;i++){

region_code[i]= rcode_begin[i] && rcode_end[i] ;

if(region_code[i]==1)

flag=0;

}

if(flag==0)

```

```

{
printf("\n Line is completely outside the window");
}

else{
slope=(float)(y1-y)/(x1-x);

if(rcode_begin[2]==0 && rcode_begin[3]==1) //left
{
y=y+(float) (W_xmin-x)*slope ;
x=W_xmin;
}

if(rcode_begin[2]==1 && rcode_begin[3]==0) // right
{
y=y+(float) (W_xmax-x)*slope ;
x=W_xmax;
}

if(rcode_begin[0]==1 && rcode_begin[1]==0) // top
{
x=x+(float) (W_ymax-y)/slope ;
y=W_ymax;
}

if(rcode_begin[0]==0 && rcode_begin[1]==1) // bottom
{
x=x+(float) (W_ymin-y)/slope ;
y=W_ymin;
}
}

```

```

// end points

if(rcode_end[2]==0 && rcode_end[3]==1) //left
{
y1=y1+(float) (W_xmin-x1)*slope ;
x1=W_xmin;
}

if(rcode_end[2]==1 && rcode_end[3]==0)    // right
{
y1=y1+(float) (W_xmax-x1)*slope ;
x1=W_xmax;
}

if(rcode_end[0]==1 && rcode_end[1]==0)    // top
{
x1=x1+(float) (W_ymax-y1)/slope ;
y1=W_ymax;
}

if(rcode_end[0]==0 && rcode_end[1]==1)    // bottom
{
x1=x1+(float) (W_ymin-y1)/slope ;
y1=W_ymin;
}

}

}

delay(1000);

clearviewport();

rectangle(W_xmin,W_ymin,W_xmax,W_ymax);

```

```
    line(0,0,600,0);  
    line(0,0,0,600);  
    setcolor(RED);  
    line(x,y,x1,y1);  
    getch();  
    closegraph();  
    return 0;  
}
```

Conclusion

- What did u learn?